Beginner's Guide to PLC Programming

by Neal Babcock





How to Program a PLC (Programmable Logic Controller)

Beginner's Guide To PLC Programming

How to Program a PLC (Programmable Logic Controller)

By Neal Babcock

Industrial Automation Series

engineer-and-technician.com

Table of Contents

Introduction to PLCs....1
Ladder Logic....1
The Dialect of PLCs....2
Equivalent Logic....4
Scan Time....6
The Automated Drill Press....7
Sequence of Operation....8
Operator Station....10
I/O Listing....11
Inputs....11
Outputs....12
Internal Coils....12
The Program....13
Machine Safeties....13
Pilot Light Test....15
Indicate The System Is Operational....16
Machine Operation Mode....16
Run The Spindle Drive Motor....17
Indicate The Spindle Drive Is Running....17
Run The Infeed Conveyor....17
Ensure There Are No Parts In The Machine....19
Ensure All Components Are At Home....19
Begin The Cycle....20
Lower The Stop Gate....21
Run The Main Conveyor....21
Indicate The Part Is In Place....22
Clamp The Part In Place....22
Lower The Spindle....23
Drilling Operation Is Complete....24
Return The Spindle To Its Home Position....25
Machine Cycle Is Complete....26
Fault Detection And System Diagnostics....27
Personnel Safety Guard Door....27
Low Compressed Air Pressure....28
Motor Starter Overload....29
Latch The Motor Overload Detection....30
Indicate A Motor Overload Condition....31
Jammed Part Detection....32
Latch The Part Jammed Detection....33
Indicate A Part Jammed Condition....33
Monitor The Drill Time....34
Summarize The Fault Conditions....35
13 Marks Of A Well Written Program....36
General PLC Tips....37
The Automated Drill Press in RSLogix 500....38

Scan Time

One critical difference between a PLC program and the equivalent electrical circuit is the issue of scanning. It works like this in most PLCs:

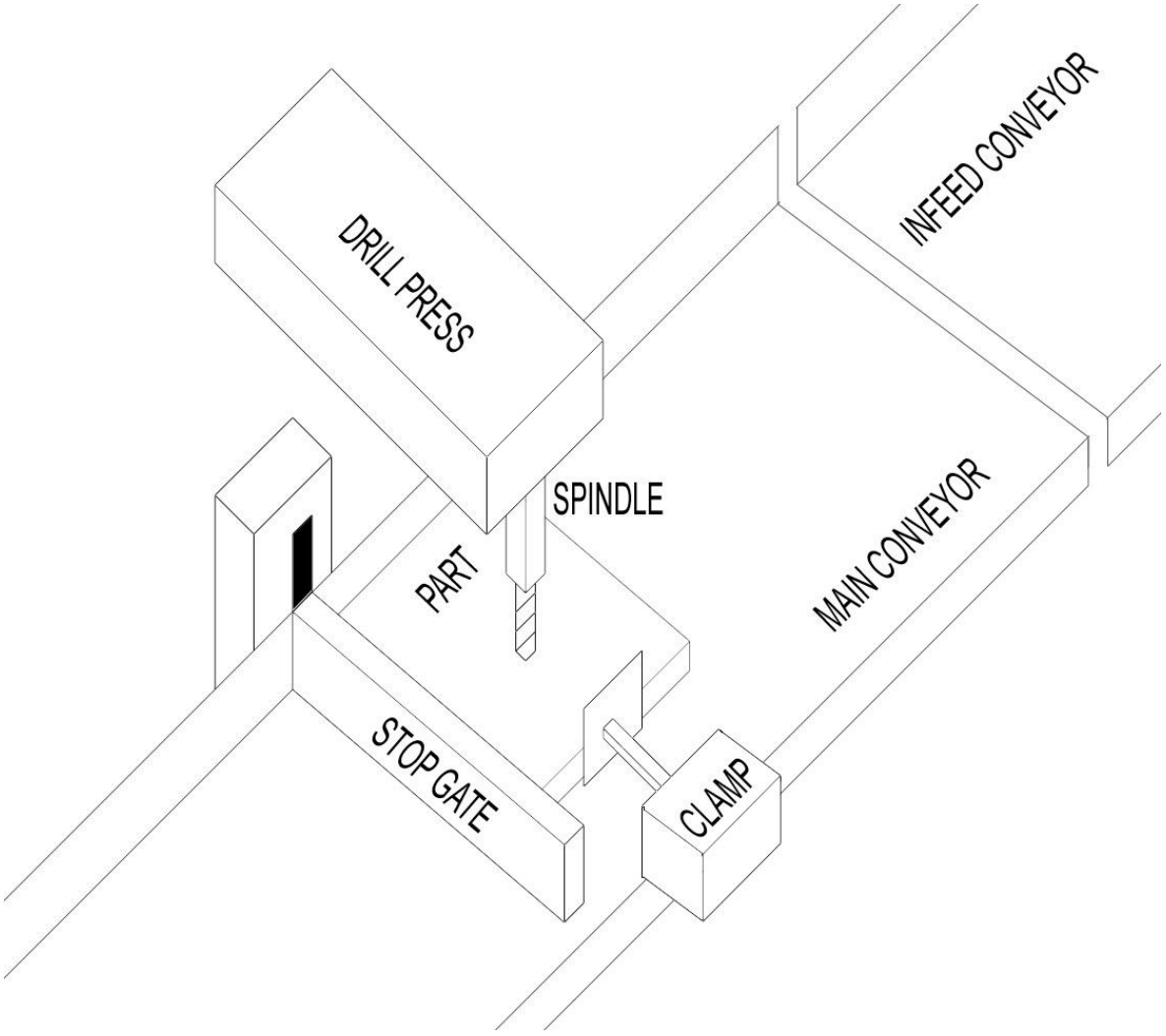
The PLC looks at the state of the inputs and stores that information in a temporary buffer. Then, it ignores what is happening electrically at the inputs. The PLC will use the information in the temporary buffer to execute the logic in the program. It will solve the logic from top to bottom, determining the truth of each rung, and turn on or turn off the appropriate addresses in the temporary buffer. When it reaches the last rung in the program, the PLC will use the data in the temporary buffer to turn on or turn off the corresponding outputs. The scan cycle is complete, and the PLC will once again look at the inputs. The amount of time this takes is called scan time, and is measured in milliseconds.

Stated more simply, the PLC reads the inputs, performs the logic and adjusts the outputs as needed.

In some newer PLCs, such as Rockwell's ControlLogix platform, it doesn't work that way. The inputs are updated during the program scan. In high-speed applications, such as bottling or pharmaceutical lines, this can cause problems.

The Automated Drill Press

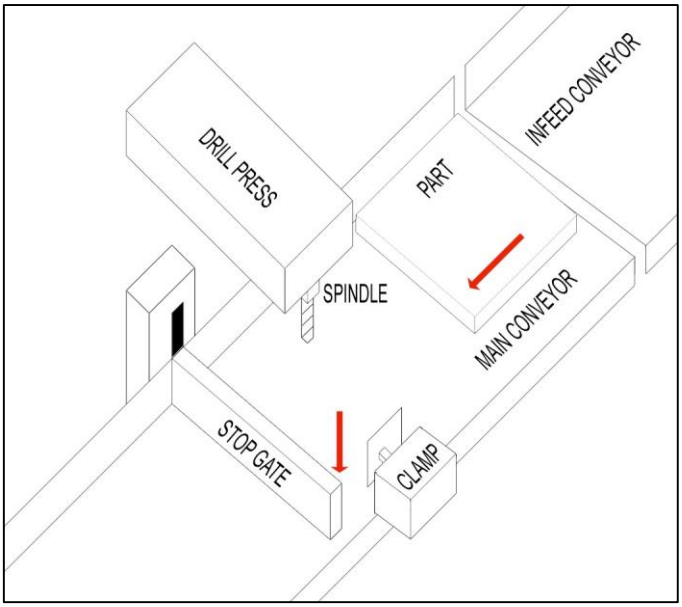
Now let's jump right into a project. The best way to learn a programming language is to look at a real world example. However, before you can do any programming, you must have a clear understanding of how the machine works. Let's say a furniture manufacturer needs to drill a 3/8" hole in a certain spot on a piece of wood. The entire process needs to be automatic. The mechanical and electrical engineers bring you an isometric drawing like the one shown here. Mechanical details have been omitted for clarity, as is often the case in a "concept" drawing.



The main conveyor will transport the part into the machine where the part will meet a pneumatically actuated stop gate. At that time, another pneumatic cylinder will actuate a clamp that will push the part back against the conveyor wall. This will hold the part in place during the drilling process. Photocells will verify that the part is in position; the spindle will lower and proceed to drill a hole in the part. After the hole has been drilled and the spindle has retracted to

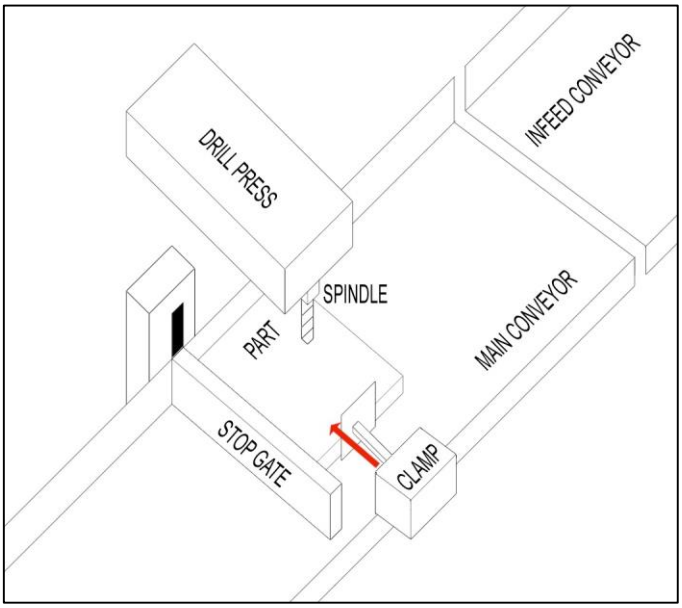
its home position, the clamp will release, the stop gate will raise and the part will exit. The cycle then repeats itself for each part that comes down the line.

Sequence of Operation



Here is a more detailed explanation of the drilling process:

When the machine starts, the stop gate lowers and the part is moved into position by the main conveyor.

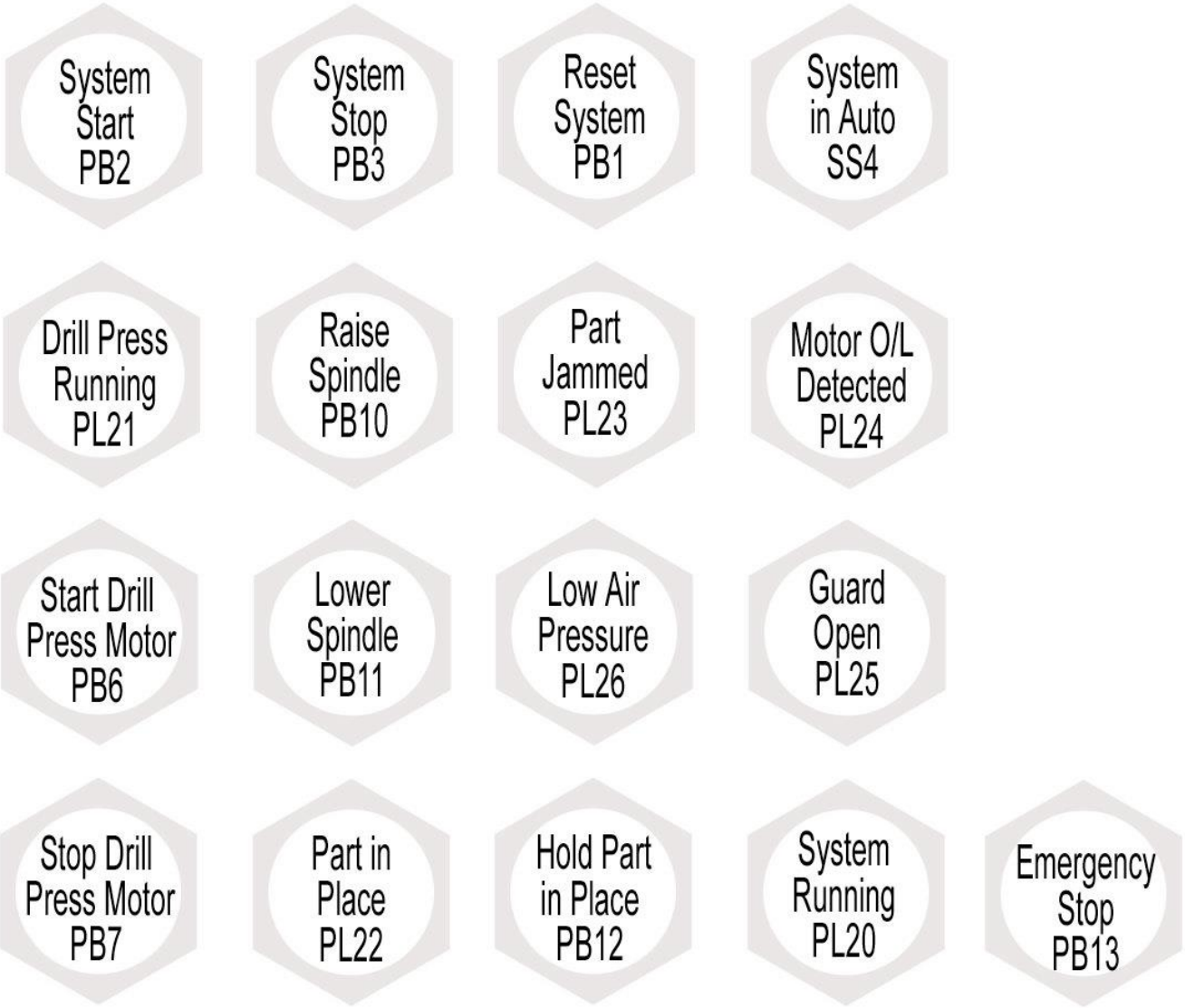


Optical sensors (photoeyes) determine when the part is in place.

When the part is positioned correctly, a clamp extends to hold the part in place.

Operator Station

An operator station for the machine, using traditional industrial switches, might look like this. Though the device name, such as PB1, may not show up on the actual station, it is a good idea to show them on your drawing.



Many machines or systems use a computerized HMI, or Human Machine Interface, to provide operator control. An HMI can be a touchscreen monitor, or a dedicated computer that is operated with a mouse, or even an Android tablet or an iPad. You can take a look at the HMI and SCADA section on engineer-and technician.com if you would like more information.

I/O Listing

It is important to define the I/O (inputs/outputs) before you begin to program. Do not skip this crucial step. Listed below is the I/O arrangement for the Automated Drill Press program.

Inputs

Address	Device Name	Device Type	Description
INPUT1	PB1	pushbutton switch	Reset System
INPUT2	PB2	pushbutton switch	Start System
INPUT3	PB3	pushbutton switch	Stop System
INPUT4	SS4	selector switch	System in Auto Mode
INPUT5	CR1	relay	Emergency Stop Cleared
INPUT6	PB6	pushbutton switch	Start Press
INPUT7	PB7	pushbutton switch	Stop Press
INPUT10	PB10	pushbutton switch	Raise Spindle
INPUT11	PB11	pushbutton switch	Lower Spindle
INPUT12	PB12	pushbutton switch	Hold Part In Place
INPUT20	MS1AUX	aux contacts on motor starter	Infeed Conveyor Running
INPUT23	MS2AUX	aux contacts on motor starter	Main Conveyor Running
INPUT24	MS5AUX	aux contacts on motor starter	Drill Press Running
INPUT26	LS26	limit switch	Guard in Place
INPUT27	PS27	air pressure switch	Air Pressure Normal
INPUT31	PSC31	photo-electric switch	Placed in X-Axis
INPUT32	PSC32	photo-electric switch	Placed in Y-Axis
INPUT33	PSC33	photo-electric switch	Part at Home
INPUT34	PSC34	photo-electric switch	Part Cleared
INPUT35	PRS35	proximity switch	Spindle Raised
INPUT36	PRS36	proximity switch	Spindle Lowered

Outputs

Address	Device Name	Device Type	Description
OUTPUT1	MS1	motor starter	Run Infeed Conveyor
OUTPUT2	MS2	motor starter	Run Main Conveyor
OUTPUT5	MS5	motor starter	Run Drill Press
OUTPUT6	DRV1	variable speed drive	Raise Spindle
OUTPUT7	DRV1	variable speed drive	Lower Spindle
OUTPUT11	SOL11	solenoid	Hold Part in Place
OUTPUT12	SOL12	solenoid	Lower Drilling Stop Gate
OUTPUT20	PL20	pilot light	System Running
OUTPUT21	PL21	pilot light	Drill Press Running
OUTPUT22	PL22	pilot light	Part in Place
OUTPUT23	PL23	pilot light	Part Jammed
OUTPUT24	PL24	pilot light	Motor Overload Detected
OUTPUT25	PL25	pilot light	Guard Open
OUTPUT26	PL26	pilot light	Low Air Pressure

Internal Coils

Address	Device Type	Description
COIL1	*(internal bit)*	System Running
COIL2	"	Pilot Light Test
COIL4	"	System in Auto Mode
COIL5	"	System in Manual Mode
COIL20	"	No Part In Machine
COIL21	"	Machine at Home
COIL22	"	Machine in Cycle
COIL23	"	Drilling Operation Complete
COIL24	"	End of Machine Cycle
COIL34	"	Part Jam Detected
COIL35	"	Excessive Drill Time Detect
COIL40	"	Motor Overload Detected
COIL50	"	System Fault

The Program

Now that you understand what has to be done and how it is to be accomplished by the mechanical equipment, you can begin writing the program.

The text in the `fixed font` is basically the information that you would see if you were looking at the monitor of the computer or a printout. In actual practice, the fonts used in PLC software vary widely. For the purposes of this book, we want to easily differentiate the program logic from our explanations of the logic.

A series of asterisks `(*****)` indicates a "rung comment." This is descriptive text added to the program, and is seen on the programming monitor, but has no affect on the logic. For purposes of this manual, I have placed additional explanations between rungs.

Use a title to name the program and include any general information.

```
|
|      ____________________________________________________________
|      |                                                           |
|      |  AUTOMATIC DRILL PRESS MACHINE CONTROL                    |
|      |  PRODUCTION LINE #3                                       |
|      |  REVISION 2                                               |
|      |___________________________________________________________|
```

Machine Safeties

It is best to start a program by evaluating any safety switches and setting a master bit. This type of bit is what we call an internal coil. It has no hardwired connection to the outside world. In this case, a latch is used to set an internal “System Running” bit. The “latch” is accomplished by putting a normally open contact around the Start System pushbutton input.

If the emergency stop is clear, and the machine guard is in place, and there is no system fault the operator may press the start button to set the latch. If the stop button is pushed or a previous conditions ceases to exist, the “System Running” latch will drop out.

Most of the time, the order of the bits in a rung doesn't matter. We could have rearranged any of the bits in this rung, though we would still have to put the latch around the Start pushbutton. The PLC wouldn't care and the output coil would still respond the same. However, to make the rung easier to read, I try to place bits from left to right in order of importance. If the E-Stop is not cleared, then nothing else should matter anyway. Having the safety guard in place is more important than a system fault. Now, if those requirements have been met, we can press the start button. And we don't care about the stop button until we have pushed the start button.

Note the instruction used for the input of PB3, the Stop System bit. It may seem backwards at first, since a hard-wired circuit would use the normally closed contacts of the switch. In fact, the

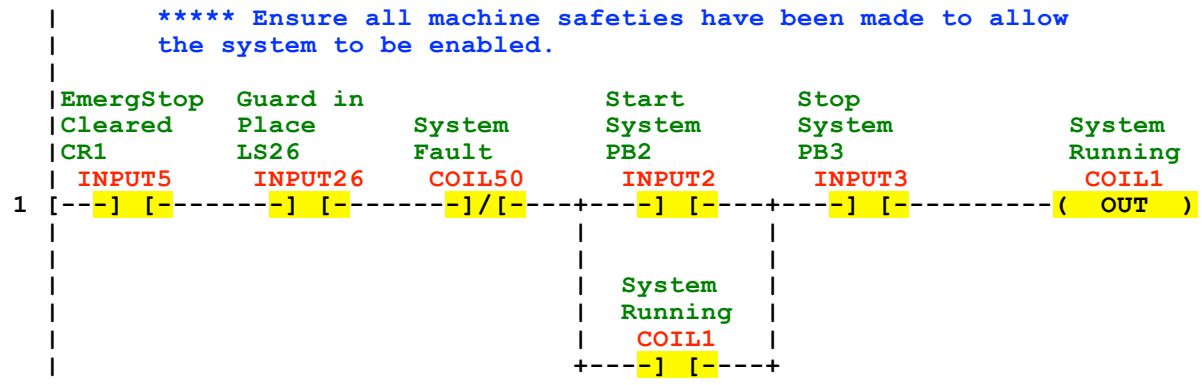
switch is wired in the failsafe position, using the normally closed contacts as the input to the PLC.

The E-Stop and the guard limit switch are called "safety interlocks." NEVER rely on a PLC to stop a machine if a contact of a safety interlock is open. ALWAYS use an interlock in a hardwired, fail-safe circuit. Use additional contacts from the switches and wire them to inputs of the PLC so that it knows the machine is to be stopped, or has stopped.

It is very important to label the bits properly. Arrange the verbs and nouns correctly. If you do, the rung will read like a sentence. You can look at this rung and say “If the Emergency Stop is Cleared and the Guard is in Place and there is no System Fault and the Start System button is pressed and the Stop System button is not pressed the System Running coil will turn on and latch itself”.

There are some simple rules that I always follow when I am writing a description for a bit:

- **Descriptions for bits require a verb.**
- **Descriptions are written to describe the “normally open” condition of the bit.**
- **The description is true when the normally open instruction of the bit is on.**
- **The description is false when the normally open instruction of the bit is off.**



The main reason for setting a master “System Running” bit is to simplify the program. For example, we don’t want the spindle motor to run if the e-stop has been pressed, or if the guard is not in place, or if there is a system fault. Rather than putting all of these bits in the rung that controls the spindle motor, we can summarize all of these bits and make the “System Running” bit. We can then place just that bit in the rung that controls the spindle motor and know that we have met all the criteria to allow the spindle motor to run.

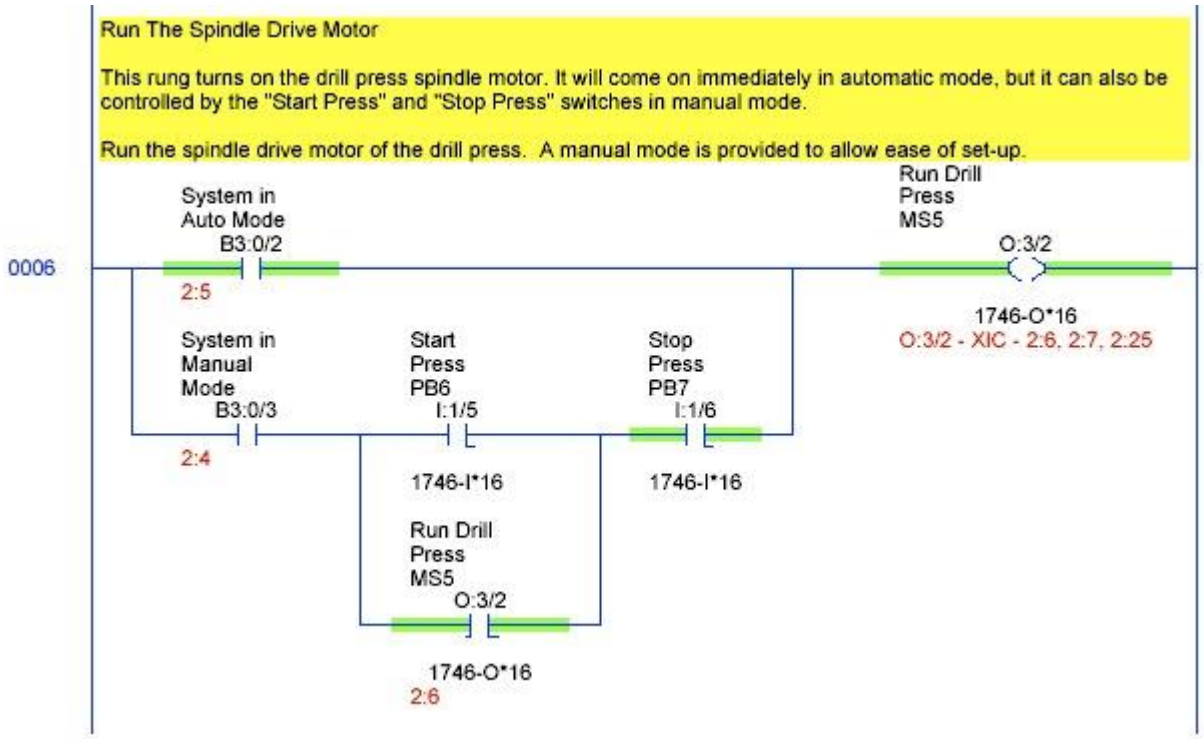
The Automated Drill Press in RSLogix 500

The most popular and most widely used manufacturer of PLCs is Rockwell Automation, who produces the Allen-Bradley MicroLogix and SLC series of PLCs. The MicroLogix and SLC families of processors and I/O modules are all programmed using Rockwell's proprietary software known as RSLogix.

I used a generic style of ladder logic in the first part of the book to explain the fundamentals of ladder logic. This style was chosen because it applies to all types of PLCs.

To give you a better understanding of how the program would look in a real world situation, I have included the same Automated Drill Press program as it would appear if it were written in Allen-Bradley's RSLogix.

Here is an example:



I will give you a brief outline of the differences between the generic format of ladder logic that I used and RSLogix's format:

First, though, the address descriptions and the rung comments are the same in the RSLogix program as they are in the generic program. Despite what the numbers say under the instruction, "System in Auto Mode" is the same bit here as the "System in Auto Mode" is in the generic program.

Outputs
Allen-Bradley labels their outputs starting with a capital “O”, whereas we used “output” in the generic program. In the rung above, “O:3/2” is the same bit as “OUTPUT5” in our generic program.

“1746-O*16” indicates the type of output module.

Inputs
Allen-Bradley labels their inputs starting with a capital “I”, whereas we used “input”. In the rung above, “I:1/5” is the same bit as “INPUT5” in our generic program.

“1746-I*16” indicates the type of input module.

Coils (internal bits)
Allen-Bradley labels *most* their internal “coils” starting with a capital “B3”, whereas we used “out”. In the rung above, “B3:0/2” is the same bit as “COIL4” in our generic program.

Cross Reference
The RSLogix program shows cross-referencing. The number “2:5” under the first instruction means that the output (or, “coil”) for that address is found in file 2, rung 5.

“2” is the first, and default, data file of a ladder in RSLogix.

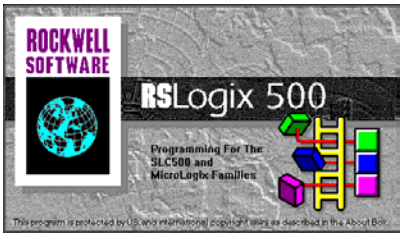
The output O:3/2 shows that its “contacts” are found on rungs 6, 7 and 25 of data file 2.

Rung Numbers
RSLogix starts with rung 0000 – the generic program starts with rung 1, so the rung numbers shown in the RSLogix program will be one less than the rung numbers in the generic program.

Logic Flow
Bits that are true in RSLogix are shown with a green highlight. This RSLogix ladder printout above is a snapshot showing the **system in automatic mode, the part is in place and the drill bit is being lowered**. There are no faults.

Other Information
RSLogix will provide a variety of information regarding the program and hardware. I have included those pages here.

RSLogix500 Project Report



Run the infeed conveyor if the machine is in automatic mode.

0008

System in Auto Mode B3:0/2 — 2:5

Run Infeed Conveyor MS1 O:3/0 — 1746-O*16 — O:3/0 - XIC -2:23

Ensure there are no parts in the machine to start the drilling cycle.

0009

Placed in X-Axis PSC31 I:2/6 — 1746-I*16

Placed in Y-Axis PSC32 I:2/7 — 1746-I*16

Part at Home PSC33 I:2/8 — 1746-I*16

Part Cleared PSC34 I:2/9 — 1746-I*16

No Part In Machine B3:0/4 — B3:0/4 - XIC -2:11

Ensure all components are at home.

0010

Hold Part in Place SOL11 O:3/5 — 1746-O*16 — 2:15

Drilling Stop Gate SOL12 O:3/6 — 1746-O*16 — 2:12

Spindle Raised PRS35 I:2/10 — 1746-I*16

Machine at Home B3:0/5 — B3:0/5 - XIC -2:11

Begin the cycle when the part has cleared the machine and all components are at home. Bit B3:0/6 will stay on during the entire drilling cycle and drop out when an end of cycle signal is generated.

0011

System in Auto Mode B3:0/2 — 2:5

No Part In Machine B3:0/4 — 2:9

Machine at Home B3:0/5 — 2:10

End of Machine Cycle B3:0/8 — 2:19

Machine in Cycle B3:0/6 — 2:11

Machine in Cycle B3:0/6 — B3:0/6 - XIC - 2:11, 2:12, 2:15 2:16, 2:17, 2:18 2:19, 2:28, 2:31

Lower the stop gate to stop the part under the spindle.

0012

Machine in Cycle B3:0/6 — 2:11

Drilling Stop Gate SOL12 O:3/6 — 1746-O*16 — O:3/6 - XIO -2:10

Run the main conveyor unless a part is being clamped.

0013

System in Auto Mode B3:0/2 — 2:5

Hold Part in Place SOL11 O:3/5 — 1746-O*16 2:15

Run Main Conveyor MS2 O:3/1 — 1746-O*16 O:3/1 - XIC -2:24

Indicate the part is in place and ready to be drilled.

0014

Part at Home PSC33 I:2/8 — 1746-I*16

Pilot Light Test B3:0/1 — 2:2

Part in Place PL22 O:3/9 — 1746-O*16

Clamp the part in place after it has reached the proper position.

0015

Machine in Cycle B3:0/6 — 2:11

Part at Home PSC33 I:2/8 — 1746-I*16

System in Manual Mode B3:0/3 — 2:4

Hold Part In Place PB12 I:1/9 — 1746-I*16

Hold Part in Place SOL11 O:3/5 — 1746-O*16 O:3/5 - XIC -2:16 XIO - 2:10, 2:13

After the part has been clamped in place, lower the spindle. The feed rate is determined by a variable speed drive.

0016

Machine in Cycle B3:0/6 — 2:11

Hold Part in Place SOL11 O:3/5 — 1746-O*16 2:15

Spindle Lowered PRS36 I:2/11 — 1746-I*16

Raise Spindle DRV1RAISE O:3/3 — 1746-O*16 2:18

Lower Spindle DRV1LOWER O:3/4 — 1746-O*16

System in Manual Mode B3:0/3 — 2:4

Lower Spindle PB11 I:1/8 — 1746-I*16